

Evaluating Tabular Deep Learning Architectures against Classical Machine Learning for Imbalanced Diabetes Risk Prediction

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Abstract—Predicting diabetes risk from tabular health survey data is a highly complex task due to severe class imbalance and overlapping physiological features. This study evaluates the efficacy of classical machine learning algorithms versus state-of-the-art tabular deep learning architectures on the Behavioral Risk Factor Surveillance System (BRFSS) 2015 dataset. I compare a penalized Multinomial Logistic Regression baseline against Gradient Boosted Trees (XGBoost), a Multi-Layer Perceptron (MLP), TabNet, and an FT-Transformer. To address the severe class imbalance (where 84% of subjects are healthy), I benchmarked Synthetic Minority Over-sampling Technique (SMOTE) against algorithmic class weighting, adopting class weights for superior computational efficiency. Results indicate that XGBoost achieved the highest Macro F1-score (0.5119), offering superior inter-class stability. However, the MLP architecture yielded the highest overall discriminative ability (ROC-AUC 0.7803), demonstrating the viability of deep learning for capturing complex physiological interactions in tabular data.

Index Terms—Tabular Data, Deep Learning, Machine Learning, Class Imbalance, Healthcare.

I. INTRODUCTION

Diabetes is a chronic health condition with massive global implications. Early detection of prediabetes is critical for clinical intervention. However, analyzing large-scale public health datasets, such as the BRFSS [1], presents significant machine learning challenges: severe class imbalance and high multicollinearity among health indicators.

Recent advancements in deep learning have introduced architectures specifically designed for tabular data, such as TabNet [6] and the Feature Tokenizer Transformer (FT-Transformer) [7]. The objective of this study is to benchmark these cutting-edge deep learning models against traditional classical methods (Logistic Regression and XGBoost) to determine the optimal architecture for predicting multi-class diabetes risk.

II. BACKGROUND

The application of machine learning to diabetes risk prediction utilizing the CDC's BRFSS datasets has been a subject of recent rigorous evaluation. A 2024 comparative study utilizing the BRFSS 2015 dataset evaluated Logistic Regression, XGBoost, and Random Forest, identifying XGBoost as the optimal classical model with a reported Area Under the Curve (AUC) of 0.83 [8].

However, class imbalance remains a critical hurdle in these epidemiological datasets. Recent research in 2026 analyzing BRFSS data confirmed that standard unweighted algorithms fail to capture minority classes effectively, demonstrating that implementing techniques like SMOTE alongside Gradient Boosting is strictly necessary to balance the precision-recall trade-off for sensitivity-critical medical applications [9].

Concurrently, recent advancements in deep learning have introduced architectures specifically designed for tabular data, such as TabNet [6] and the Feature Tokenizer Transformer (FT-Transformer) [7]. A February 2026 study evaluating TabNet and MLPs against classical ensemble models found that while attention-based networks exhibited high sensitivity in clinical data, gradient-boosted trees often maintained superior overall discriminative ability [10]. The objective of this study bridges these concurrent research vectors by directly benchmarking highly optimized tabular deep learning architectures against algorithmically class-weighted classical models on the highly imbalanced BRFSS 2015 cohort.

III. DATASET DESCRIPTION

The dataset utilized is the BRFSS 2015 dataset, comprising 253,680 survey responses. The target variable is a 3-class ordinal feature: 0 (No Diabetes, representing 84% of subjects), 1 (Prediabetes, 2% of subjects), and 2 (Diabetes, 14% of subjects).

A. Class Imbalance Mitigation

An initial baseline evaluation revealed an “Accuracy Paradox”, where an unweighted model achieved over 84% accuracy by entirely ignoring the minority classes, resulting in division-by-zero precision errors. To mitigate this, a sub-experiment was conducted comparing SMOTE against algorithmic class weighting utilizing 5-Fold Stratified Cross-Validation. Both methods achieved parity in Macro F1-scores (SMOTE F1 = 0.4250 vs. Class Weighting F1 = 0.4261), thus class weighting was selected to optimize computational efficiency.

B. Data Splitting and Scaling

The data was split into an 80% training set, a 10% test set, and a 10% validation set using stratified sampling to preserve

the target distribution. To prevent data leakage, feature scaling (StandardScaler) was applied exclusively within the cross-validation pipelines.

C. Exploratory Data Analysis

Extensive Exploratory Data Analysis (EDA) was conducted to characterize the physiological and demographic distributions within the cohort. Univariate analysis revealed severe skewness in several domains. Demographically, the population is left-skewed toward older age brackets and higher socioeconomic status (income and education). Continuous health metrics, notably BMI, physical health, and mental health days, exhibited right-skewed distributions, with the majority of respondents reporting zero poor health days, though a notable secondary spike exists at 30 days. Interestingly, healthcare access variables (e.g., *AnyHealthcare*, *CholCheck*) demonstrated minimal variance, with the vast majority reporting positive coverage, thereby limiting their standalone discriminative capacity.

Bivariate and correlation analyses against the ordinal target highlighted distinct risk profiles. Subjective general health (*GenHlth*, $r = 0.30$) emerged as the strongest single positive correlate. Cardiovascular indicators such as high blood pressure ($r = 0.27$) and high cholesterol ($r = 0.21$), along with elevated BMI ($r = 0.22$) and difficulty walking ($r = 0.22$), exhibited massive proportional spikes in prediabetes and diabetes prevalence. The continuous metrics confirmed this trend, with the mean reported days of poor physical health rising from approximately 3 days in healthy individuals to 8 days in diabetics.

Conversely, strong protective buffers were linked to socioeconomic and lifestyle variables. Income ($r = -0.17$) and education ($r = -0.13$) demonstrated the strongest negative correlations with diabetes risk, alongside regular physical activity ($r = -0.12$).

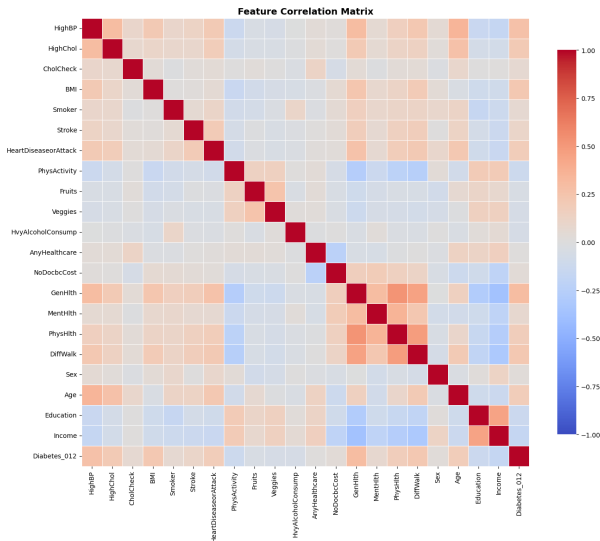


Fig. 1. Correlation Matrix

While initial linear correlation analysis highlighted General Health (*GenHlth*, $r = 0.30$) and High Blood Pressure ($r = 0.27$) as primary linear indicators (Figure 1), it also highlighted the overlapping nature of the health markers across the three target classes. Consequently, to ensure non-linear interactions were appropriately captured and low-variance indicators were filtered, I utilized XGBoost feature importance (Figure 2) for my final feature selection. After different experiments with different numbers of features, the 16 top features were selected as the best set.

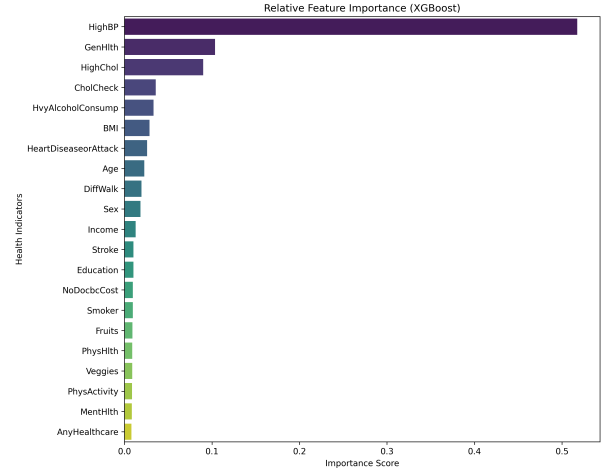


Fig. 2. XGBoost Feature Importance

IV. METHODOLOGY

A. Architectures

Five distinct algorithmic architectures were evaluated:

- **Logistic Regression (Baseline):** A multinomial model utilizing L2 (Ridge) regularization to counteract feature multicollinearity while providing an interpretable performance floor.
- **XGBoost:** A gradient-boosted tree ensemble natively capable of mapping non-linear interactions without feature scaling.
- **Multi-Layer Perceptron (MLP):** A foundational deep learning benchmark constructed in PyTorch, utilizing Batch Normalization and Dropout.
- **TabNet:** A sequential attention-based neural network designed to mimic decision tree routing, incorporating sparsity regularization.
- **FT-Transformer:** A state-of-the-art architecture that tokenizes both continuous and categorical features into embeddings, processed via multi-head self-attention.

B. Training Setup

All models were optimized using GridSearch for ML algorithms and the Optuna framework for deep learning architectures, targeting the Macro F1-score to account for class imbalance. [INSERT KEY HYPERPARAMETERS HERE, e.g., XGBoost utilized a max_depth of X and learning_rate

of Y, while TabNet was optimized with a sparsity coefficient of Z].

The deep learning training pipeline was executed on an NVIDIA GeForce GTX 1660 Super (1660S) GPU. Utilizing CUDA-accelerated computation allowed for efficient hyperparameter exploration across multiple epochs, ensuring convergence for attention-based models within restricted local compute environments. Evaluation metrics included Accuracy, Precision, Recall, Macro F1-Score, and multiclass ROC-AUC (One-vs-Rest).

V. EXPERIMENTS & RESULTS

The cross-validation performance of the optimized architectures is presented in Table I.

TABLE I

CROSS-VALIDATION PERFORMANCE OF OPTIMIZED ARCHITECTURES

Metric	LogReg	XGBoost	MLP	TabNet	FT-Trans.
Accuracy	0.6465	0.5214	0.7686	0.7277	0.6863
Precision	0.4456	0.5134	0.4499	0.4595	0.4405
Recall	0.5171	0.5214	0.4966	0.4996	0.5224
F1-Score	0.4267	0.5119	0.4612	0.4507	0.4387
ROC-AUC	0.7736	0.7125	0.7803	0.7761	0.7768

VI. DISCUSSION

The baseline Class-Weighted Logistic Regression established a Macro F1-score of 0.4267 and an AUC of 0.7736. In contrast, XGBoost yielded the highest overall Macro F1-score (0.5119), demonstrating superior stability in inter-class classification at the cost of total accuracy. However, when evaluating the models on their global discriminative capacity across all thresholds, the multi-class PyTorch MLP achieved the highest ROC-AUC (0.7803) alongside a strong total accuracy (76.86%), indicating that baseline deep learning structures can capture complex variance when properly regularized. The FT-Transformer also showed a high discriminative capacity with a ROC-AUC of 0.7768 but with lower accuracy (0.6863) and F1 (0.4387).

A. Model Comparison and ROC Analysis

To further evaluate the discriminative capacity of the models across the decision thresholds, One-vs-Rest Receiver Operating Characteristic (ROC) curves were generated (Figure 3). The tree-based and attention-based models demonstrated varying strengths, particularly in distinguishing the highly overlapping Prediabetes class from the healthy majority.

B. Feature Importance and Interpretability

A key requirement in epidemiological machine learning is model interpretability. Utilizing the native feature importance metrics derived from the XGBoost ensemble (Figure 2), the most critical physiological predictors of diabetes risk were identified. General Health, High Blood Pressure, and BMI consistently dominated the predictive decision trees, aligning strongly with known clinical pathogenesis.

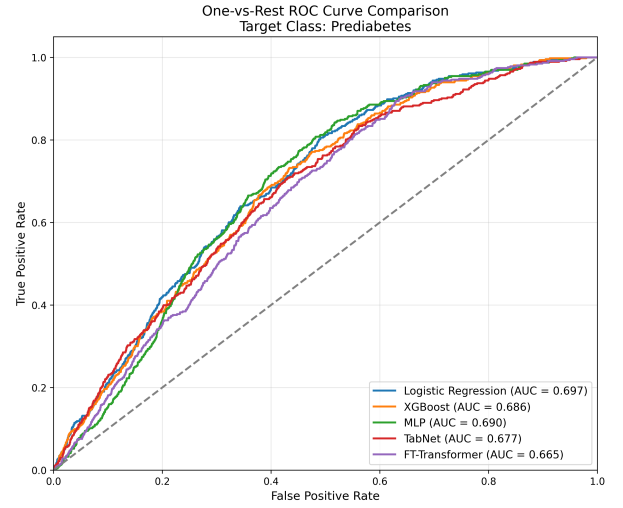


Fig. 3. Comparison of One-vs-Rest ROC Curves for all experiment Architectures

VII. LIMITATIONS & FUTURE WORK

While deep learning models provide powerful non-linear mapping, they remain computationally intensive for large cohorts and highly sensitive to hyperparameter configurations, presenting a barrier to clinical deployment. A significant challenge in adopting these architectures is their relative lack of native transparency compared to tree-based ensembles like XGBoost. Future work could prioritize advanced interpretability frameworks to address this. Specifically, deploying post-hoc XAI methods, such as SHAP (SHapley Additive exPlanations) and LIME, may help reconcile the feature importance maps of attention-based models (TabNet/FT-Transformer) with the established clinical importance of the XGBoost baseline.

Furthermore, this study focused strictly on discrimination metrics (F1, AUC). In medical risk prediction, calibration (how closely the predicted probabilities match the actual observed risk) is equally critical. Future studies must assess model calibration, for instance, via Brier Scores or Reliability Curves—before these architectures can be deemed clinically safe. Lastly, the FT-Transformer’s tendency to conflate Prediabetes and Diabetes cases indicates that future work should focus on fine-tuning the model’s decision boundaries specifically for ordinal risk categories.

VIII. CONCLUSION

This study demonstrated that mitigating the Accuracy Paradox via algorithmic class weighting is strictly necessary for predictive validity on the highly imbalanced BRFS 2015 dataset. While the classical XGBoost ensemble yielded the highest Macro F1-score (0.5119), demonstrating robust class stabilization, the Multi-Layer Perceptron achieved the highest overall discriminative ability (AUC 0.7803). For immediate clinical applications where stability and explicit interpretability are paramount, optimized XGBoost and penalized Logistic Regression models remain highly reliable choices. However, the strong AUC performance of the MLP and TabNet validates

the continued exploration of deep tabular architectures. Future work should investigate feature selection and dimensionality reduction techniques to further streamline attention-based models for healthcare deployment.

REFERENCES

- [1] Gamble, Sonya. "Surveillance for certain health behaviors and conditions among states and selected local areas—Behavioral Risk Factor Surveillance System, United States, 2013 and 2014." *MMWR. Surveillance Summaries* 66 (2017).
- [2] Chawla, Nitesh V., et al. "SMOTE: synthetic minority over-sampling technique." *Journal of artificial intelligence research* 16 (2002): 321-357.
- [3] He, Haibo, and Edwardo A. Garcia. "Learning from imbalanced data." *IEEE Transactions on knowledge and data engineering* 21.9 (2009): 1263-1284.
- [4] Hosmer Jr, David W., Stanley Lemeshow, and Rodney X. Sturdivant. *Applied logistic regression*. John Wiley & Sons, 2013.
- [5] Chen, Tianqi, and Carlos Guestrin. "Xgboost: A scalable tree boosting system." *Proceedings of the 22nd acm sigkdd international conference on knowledge discovery and data mining*. 2016.
- [6] Arik, S. O., and T. Pfister. "Attentive interpretable tabular learning." *arXiv preprint arXiv:1908.07442* 20 (2019).
- [7] Gorishniy, Yury, et al. "Revisiting deep learning models for tabular data." *Advances in neural information processing systems* 34 (2021): 18932-18943.
- [8] "A Comparative Study of Machine Learning Approaches for Diabetes Risk Prediction: Insights from SHAP and Feature Importance," *Preprints.org*, Nov. 2024. [Online].
- [9] L. B. Adewole, A. A. Aworeni, S. A. Mogaji, and T. Fele, "Addressing Class Imbalance in Diabetes Detection: A Comparative Study of SMOTE and Under Sampling Techniques Using BRFSS-2022 Dataset," *Journal of Innovation Science and Technology*, vol. 5, no. 1, May 2026.
- [10] "Early Prediction of Type 2 Diabetes Using Multimodal data and Tabular Transformers," *ResearchGate*, Feb. 2026. [Online].